I/O Products

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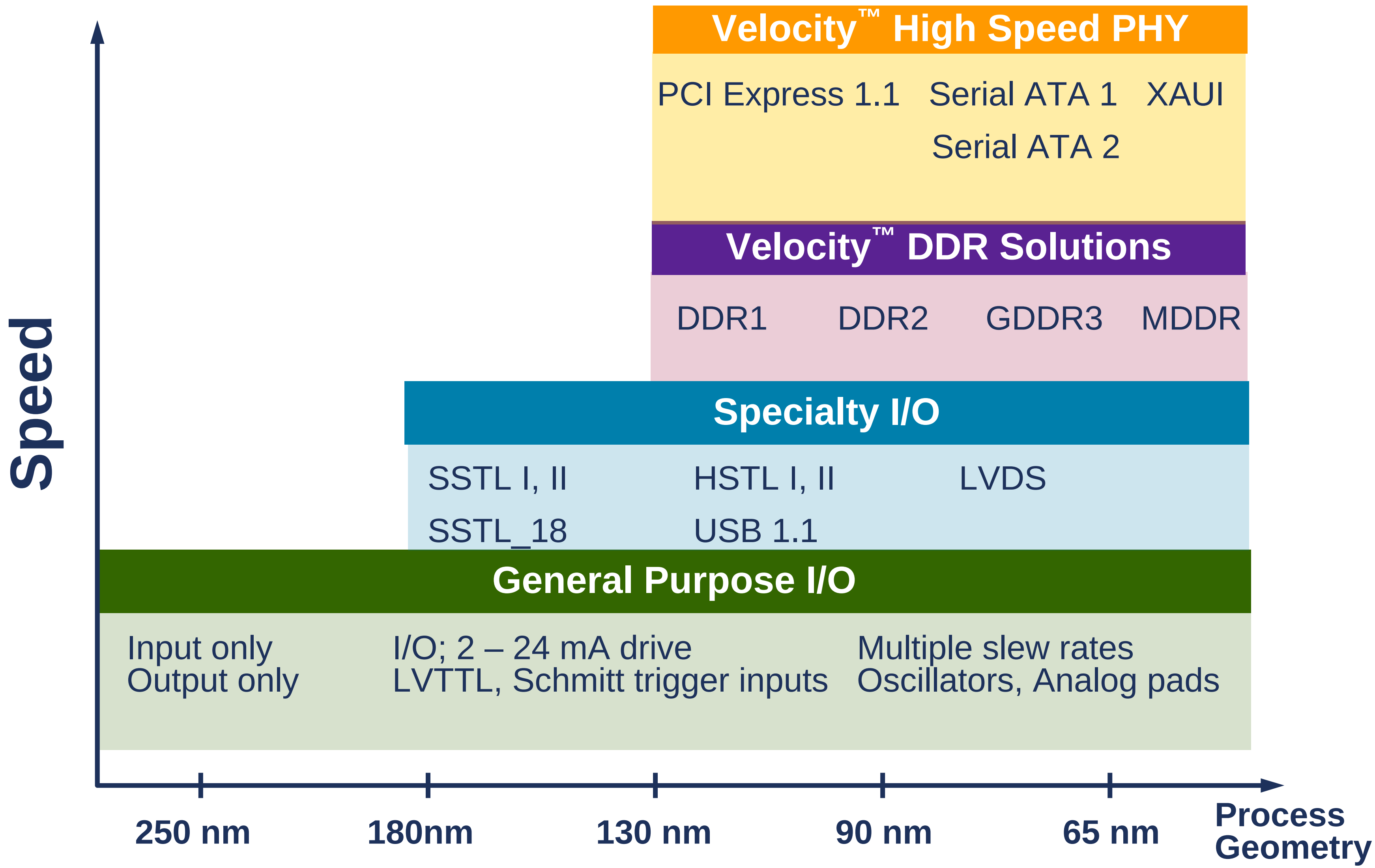
James Lee

I/O Products Complete the Story



I/O Product families

- **General Purpose I/O**
  - Generally low speed (<100MHz)
    - Input only, Output only, Bi-Directional pads
  - Pad pitch and configuration dictated by specific design needs
    - In-line pad ring, staggered pad ring, wire bond, flip chip
  - Analog and Crystal Oscillator ‘I/O’ cells usually included
    - Can integrate these cells into the pad ring
- **Specialty I/O**
  - Mostly standards-based for interface applications
    - PCI, PCI-X
  - Memory Interface applications
    - SSTL2, HSTL
  - High-Speed data transfer applications
    - LVDS (666MHz to 1GHz)

ARM GPIO Physical IP Availability

Foundry	250nm	180 -150nm	130 -110nm	90 - 80nm	65nm
Chartered		✓	✓	✓	✓
IBM		✓	✓	✓	✓
TSMC	✓	✓ T,N	✓ T,N	✓ T,N	Coming Soon
UMC	✓ N		✓		
1st Silicon	✓	✓			
Dongbu Anam		✓			
MagnaChip					
SMIC					
Silterra		✓			
HHNEC		✓			
HeJian		✓			
Tower		✓	✓		
Grace		✓			
Vanguard					

Free Library Program **Blue = In Development** **N = NGPIO** **T = TSMC designed**

Specialty I/O Product Choices

- Designer’s Challenge - Chip Interfaces
  - As performance increases, chip boundaries are becoming the data bandwidth bottleneck
  - Customers need access to a variety of interface standards that address board level integration issues

ARM Specialty I/Os	Application Examples
HSTL	Memory Interface: SRAM
SSTL 18, 2 & 3	Memory Interface: SDRAM
LVDS (622-1.2GHz)	High Speed Communications
PCI & PCI X	PCI Interface

Specialty I/Os for license to End Users

I/O Products	Application	Top Speed	TSMC Availability
SSTL18	SDRAM Interface: DDR Memory FCRAM	350MHz (Class I)	90nm Class I
SSTL2	SDRAM Interface: DDR Memory FCRAM	Up to 267MHz (Class I) Up to 333MHz (Class II)	180nm Class I/II 130nm Class I/II 90nm Class I/II
HSTL	SRAM Interface: QDR Memory DDR Memory RLDRAM	Up to 350MHz (Class I) Up to 350MHz (Class II)	130nm Class I/II 90nm Class I/II
PCI-66	PCI Interface	Up to 66MHz	180nm 130nm
PCI-X	PCI Interface	Up to 266MHz	180nm 130nm
LVDS	Network Interface High Speed - Communication	Up to 850MHz	180nm 130nm 90nm

ARM SPIO Physical IP Availability

Foundry	250nm	180nm	130nm	90nm	65nm
Chartered				HSTL, SSTL18, SSTL2	HSTL, SSTL18, SSTL2
IBM				HSTL, SSTL18, SSTL2	HSTL, SSTL18, SSTL2
TSMC		USB1.1, PCI-X, LVDS, SSTL2	HSTL, LVDS, PCI, PCIX, SSTL2	HSTL, LVDS, SSTL18, SSTL2	Coming Soon
UMC				HSTL, LVDS, PCIX, PECL, SSTL2, USB1.1, USB2.0	
1st Silicon					
Dongbu Anam					
MagnaChip					
SMIC					
Silterra					
HHNEC					
HeJian					
Tower			HSTL, LVDS, PCIX, USB1.1		
Grace					
Vanguard					

Free Library Program

Blue = In Development

DDR Products

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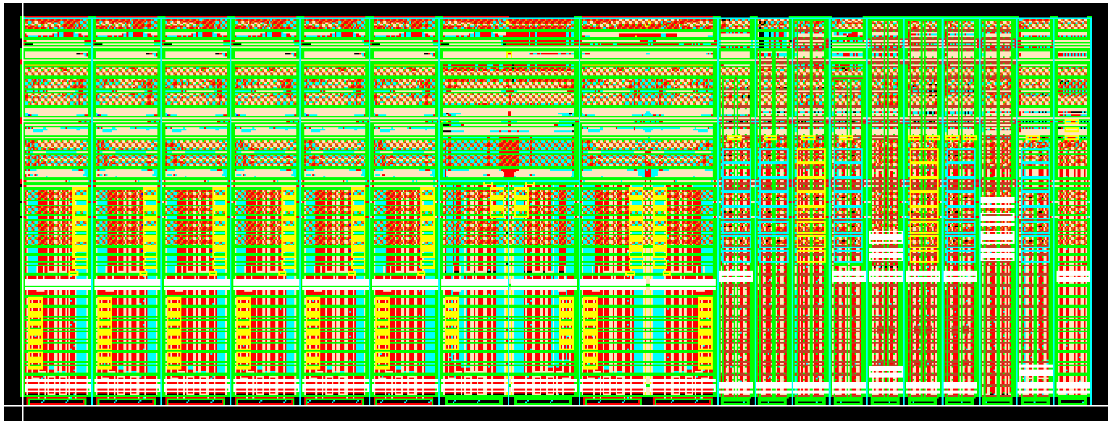
Wide Variety of DDR Applications

Mainstream devices moving towards high performance

Cell phone #1 New Market for DDR

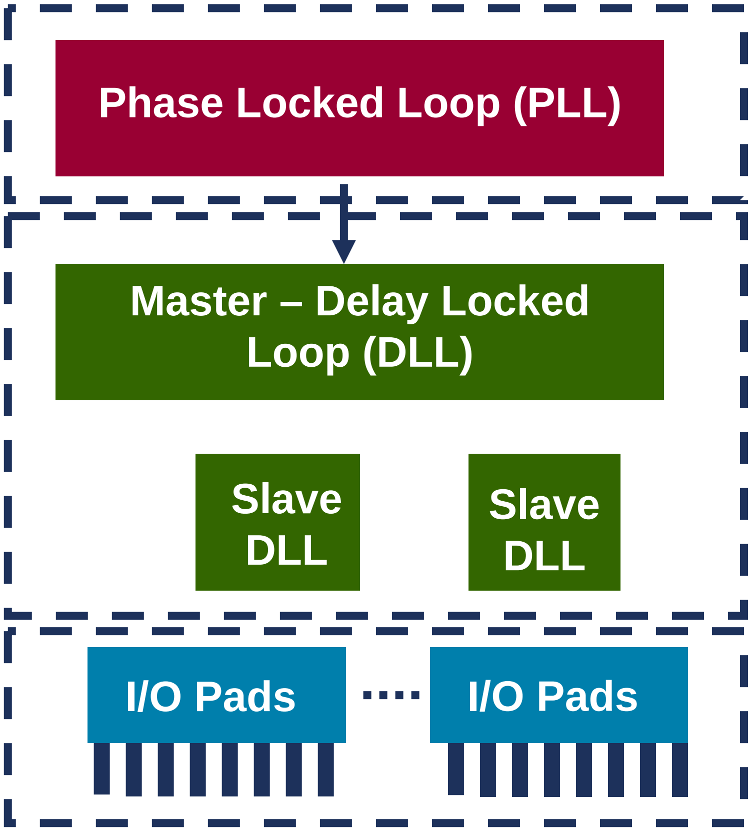
DDR Silicon PHY IP Components

- IO cell set
  - Combo(DDR1/2) or dedicated set (MDDR, GDDR3)
  - Power / ground
  - Decap cells
- DLL master & slave
- Deskew PLL
- ARM Fabric DDR controllers (ARM Mobile DDR PL340)



DDR Combo pad set

Flexible DDR Solution

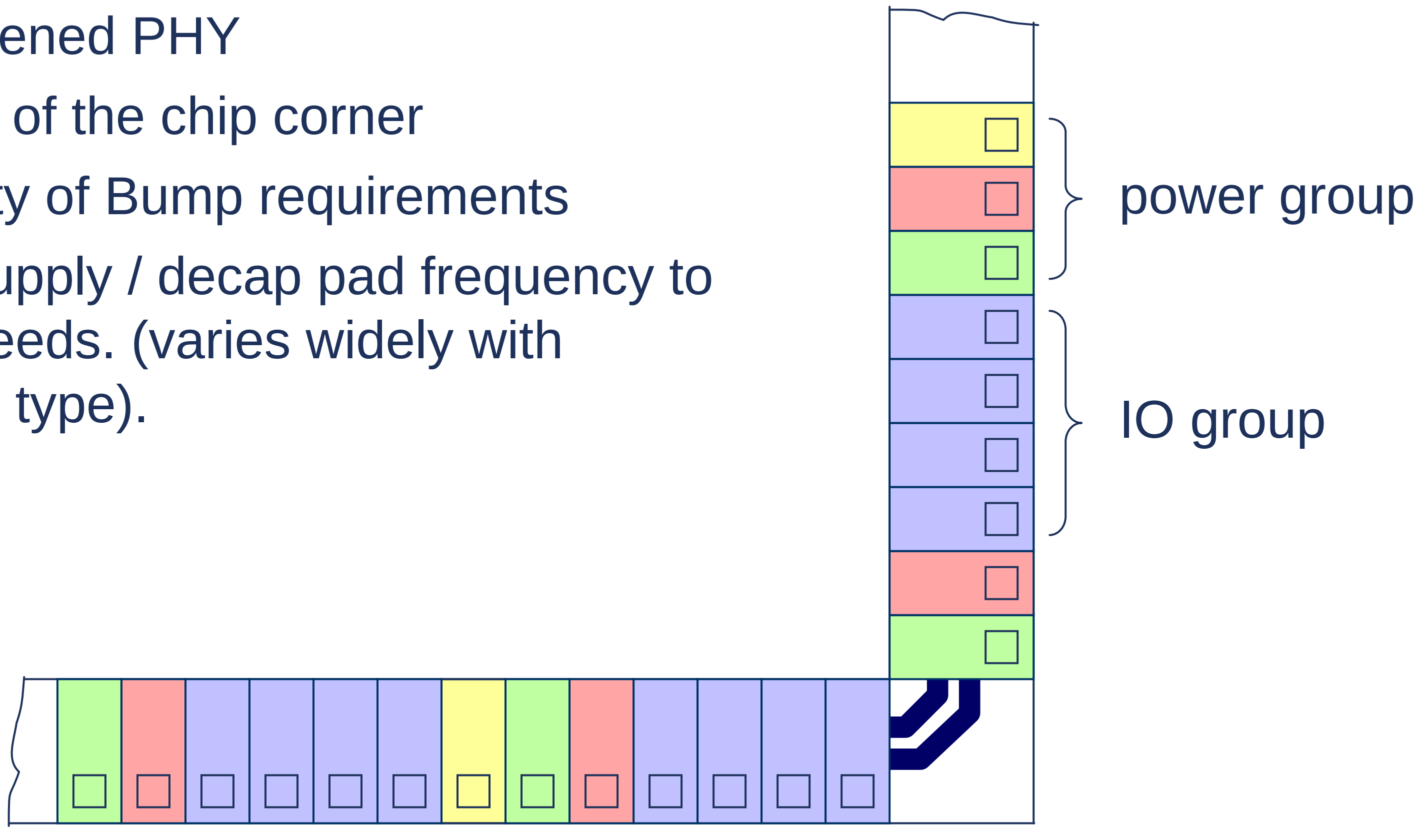


Off Chip Interface

- Flexible implementation to optimize solution for:
  - -area
  - -power
  - -performance
- Switchable architecture allows DDRI, DDRII, GDDR3
- Backwards compatible with DDR (2.5V I/O drive) using lower drive strengths
- Cost savings:
  - -Available in 2 or 3 piece solutions

IO Placement Flexibility

- No Hardened PHY
- Problem of the chip corner
- Variability of Bump requirements
- Match supply / decap pad frequency to actual needs. (varies widely with package type).



DDR Solutions for 90nm

Standard/App	Soft IP: Controller	Hard IP: Analog	Hard IP: I/Os
DDR I 2.5V **SDRAM** *Up To 400Mb/s*	ARM PL340	800Mbs DLL 533MHz PLL	DDR1/2
DDR II 1.8V **SDRAM/ RLDRAM** *Up To 800Mb/s*	3rd Party	800Mbs DLL 533MHz PLL	DDR1/2
GDDR III 1.8V **SDRAM/ RLDRAM** *Up To 1.6Gb/s*	3rd Party	1800Mbs DLL 1800MHz PLL	GDDR3

DDR Solutions for 130nm

Standard/App	Soft IP Controller	Hard IP Analog	Hard IP I/Os
DDR I 2.5V — SDRAM *Up To 400Mb/s*	ARM PL340	800Mbs DLL 533MHz PLL	DDR1/2 MDDR
DDR II 1.8V — SDRAM/ RLDRAM *Up To 800Mb/s*	3rd Party	800Mbs DLL 533MHz PLL	DDR1/2
GDDR III 1.8V — SDRAM/ RLDRAM *Up To 1.4Gb/s*	3rd Party	1400Mbs DLL 800MHz PLL	DDR1/2/GDDR3

Our DDR PHY Customers

3Dlabs

Xambala™
The Semantic Processor Company

Chelsio Communications
Accelerate

CRADLE TECHNOLOGIES

LSI LOGIC

Open-Silicon

tsmc

UBICOM

Silverback Systems
Accelerating IP Storage Networks

AZUL SYSTEMS

eSilicon

TERARECON, INC.

sandburst

ageia

Agilent Technologies

BROADLOGIC

NORDIC SEMICONDUCTOR